Emergent Tonal Inference: Bridging Triadic Models and Melodic Reality

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Abstract

Human listeners effortlessly process both vertical harmony and horizontal melody when they hear music, yet computational models often struggle to bridge these two different forms. Many connectionist models rely on vertical triadic representations, which work well for static chords but often struggle when applied to more complex sequential melodies. In this study, we tested the limits of a simple feed-forward multilayer perceptron, trained with synthetic chords, by exposing it to real-world musical input. To process sequential audio, we implemented an Exponential Moving Average (EMA) buffer to simulate human short-term memory. We then evaluated the model on the two famous musical excerpts, “Twinkle Twinkle Little Star” (Simple, Monophonic) and “Twelve Variations on "Ah vous dirai-je, Maman"” (Complex, Ornamented). Our results show that forcing rapid sequential melodies into vertical chord templates causes substantial instability in tonal-center estimation, resulting in hallucinated key prediction. In contrast, our baseline control method “pitch-class accumulation” successfully captured the global key of both pieces. These findings suggest that tonal instability originates primarily from representational mismatch rather than noisy auditory input, which reveals an important trade-off between sensitivity to local harmonic structure and tolerance to sequential melodic variation.

Keywords: music cognition, tonal inference, harmony, melody, pitch-class, connectionist modeling, tonal center, representation

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1. Introduction

1.1. The Harmony and The Melody

When human listeners hear music, they are capable of integrating two fundamentally distinct structural forms: Harmony and Melody. Harmony is vertical; it presents a high density of acoustic information simultaneously at a single point in time, and allows listeners to immediately extract a tonal center. On the other hand, melody is horizontal; it presents sequential information and spreads across a temporal window, requiring our cognitive system to hold previous notes in short-term auditory memory to infer the underlying harmonic structure. While human cognition is able to bridge these two different dimensions, computational modeling of music perception often struggles to properly differentiate them.

1.2. The Representational Problem

Many computational models of tonal inference assume that music can be effectively reduced into discrete and vertical chord units For instance, classical connectionist frameworks usually train feed-forward neural networks to map isolated triads onto specific musical keys (Bharucha & Todd, 1989). While these models successfully demonstrate emergent statistical learning under ideal harmonic conditions, they struggle with the critical representational mismatch introduced by real-world sequential melodies. If a cognitive model is forced to interpret rapid sequential single notes as vertical chords, it becomes vulnerable to systematic failure, especially when the melody contains dense chromatic ornamentation.

1.3. Our study & Models

This paper investigates the representational boundary between vertical harmonic models and horizontal melodic sequences. We utilized a feed-forward multilayer perceptron originally trained on synthetic diatonic chords to process real-world audio. In addition to the base model, we implemented an Exponential Moving Average (EMA) buffer to simulate human echoic memory. This additional step allowed us to test the network's tonal inference capabilities on both simple monophonic melodies (*Twinkle Twinkle Little Star*) and their highly ornamented sequences (*Mozart's 12 Variations*). Finally, we compared the network's triadic-forcing performance against a baseline pitch-class accumulation method to access whether tonal collapse originates from noisy auditory data or from the structural limitations of the representation method itself.

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2. Methods

2.1. The computational Architecture

The foundation of our pipeline relies on a feed-forward multilayer perceptron which was originally developed to map musical chords to their corresponding tonal centers. Instead of relying on hard-coded rules of Western harmony, our network is designed to learn tonal structures through exposure to statistical regularities, following the connectionist framework by Bharucha and Todd (1989).

For the machine-learning part, we trained the model exclusively on perfect synthetic diatonic chords. We purposely exposed the network to the isolated chord structures for it to learn the underlying statistical pattern of tonal spaces autonomously. During inference, the network takes an input vector representing the current chord, then processes it through a hidden layer, and finally produces a probability distribution across all 12 major + 12 minor keys. This process establishes a vertical and harmonic baseline for the network, as we choose to strictly optimize the network to infer a tonal center from uninterrupted and simultaneously occurring notes.

2.2. The Audio Translation Pipeline & Temporal Memory

To evaluate the model on real-world music containing both harmonic and sequential structure, we developed a pipeline to bridge the continuous audio data with the discrete input requirements of the neural network. We sourced raw MIDI files from online, then segmented the audio into discrete temporal windows and extracted chroma vectors (pitch-class profiles) for each audio frame.

However, human auditory perception does not process sound and music notes individually. Rather, our brain holds onto recent acoustic information to bridge the temporal gaps between “each” sound we received. This action allows us to bridge the information continuously until the input stops. If the network were asked to process each temporal slice independently, the rapid changing melodic input would result in highly unstable and noisy input vectors. To resolve this issue, we implemented an Exponential Moving Average (EMA) memory buffer. This buffer is mathematically defined as:

$$C_t = \alpha \cdot C_{t-1} + (1 - \alpha) \cdot X_t$$

Where C_t represents the smoothed cognitive representation of the audio at time t, X_t is the auditory input at time t, and α determines the decay rate of the auditory memory. With the help of this memory buffer, we were able to mimic human echoic memory and provide the necessary sequential context to infer tonal centers over time for the neural network.

2.3. Representational Forcing (Triadic vs. Pitch-Class)

To evaluate how our input structure affects the tonal inference, we processed the temporally smoothed audio through two different representational constraints before evaluating tonal inference:

- **Method A: Triadic Forcing (The Experimental Condition).**

  This method tests the original model's assumption that music is composed of discrete and vertical chord units. The smoothed chroma vector is compressed into the closest major or minor triad (3 notes). This forced triad approximation is then passed into the neural network for musical key prediction, which results in a strictly harmonic interpretation of the feed-in audio sequence.

- **Method B: Pitch-Class Distribution (The Control Baseline).**

  This method serves as the baseline control condition that entirely bypasses both chord reconstruction and the neural network, which helped us to confirm whether any systematic failures are caused by the audio itself or Method A. This approach accumulates raw note frequencies (pitch classes) over time across the duration of the musical excerpt. We then compared the resulting distribution directly against fixed global key templates (For example, a C Major scale profile) using straight statistical matching. On the other hand, Method B preserves global pitch distribution while discarding local harmonic sequencing.

3.1. Baseline Validation

Before evaluating the model on real-world audio file, we first needed to validate the network's baseline capacity to track tonal shifts using synthetic harmonic input. To do this, we fed the model a synthetic chord progression specifically designed to modulate from C Major to G Major. This sequence consisted of an I-V-vi-IV pattern in C major, followed by a corresponding progression centered around G major:

Chord index:

C:maj | *G:maj* | *A:min* | *F:maj* | *C:maj* | *G:maj* | *A:min* | *F:maj* | *C:maj* | *G:maj* | *A:min* | *F:maj* | *G:maj* | *D:maj* | *E:min* | *C:maj* | *G:maj* | *D:maj* | *E:min* | *C:maj* | *G:maj* | *D:maj* | *E:min* | *C:maj*

The results demonstrated correct tonal tracking when given clean and vertical harmonic input. As shown in Figure 2, the model's probability estimates updated dynamically across the chord sequence as it analyzes the probability distribution across the chord sequence index. Around the midpoint of the sequence (chord index = 12), the probability of C Major decreased, while G Major and its relative E Minor spiked together.

The Circle of Fifths visualization also supports our interpretation. As shown in Figure 3, we can see the model mapped a clear trajectory from C Major region toward G Major. Together, these results confirm that the network and visualization pipelines are fully functional under ideal harmonic conditions. When the input is consistent with the model's training criteria, the model successfully infers and tracks dynamic tonal centers.

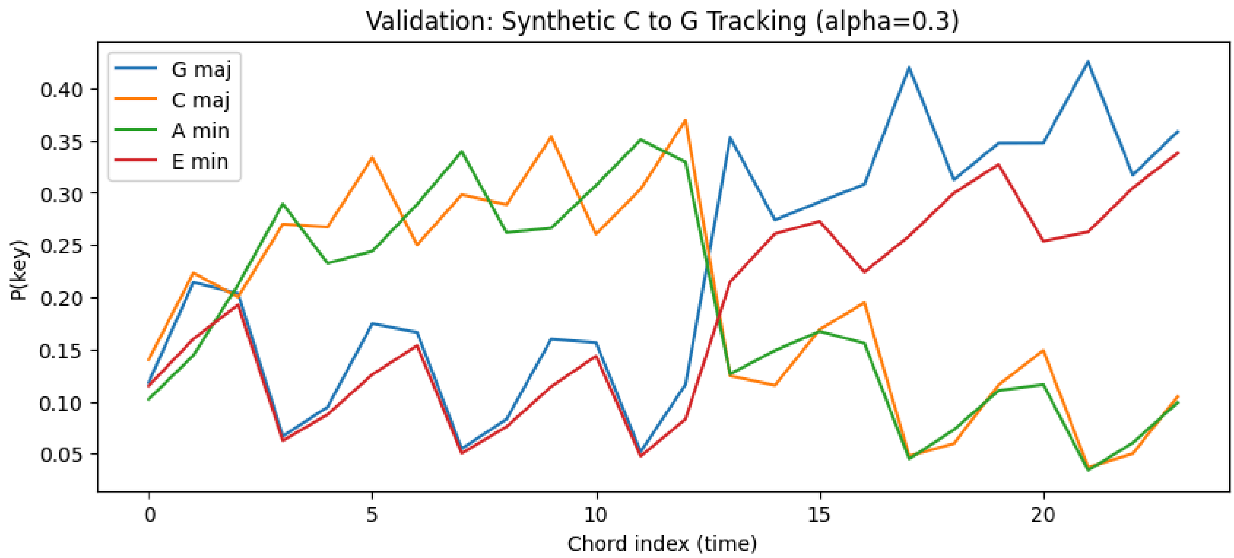

Figure 2. Validation: Synthetic C to G Tracking. Key probability estimates across a synthetic chord sequence designed to modulate from C major to G major. The x-axis represents chord index, not real time.

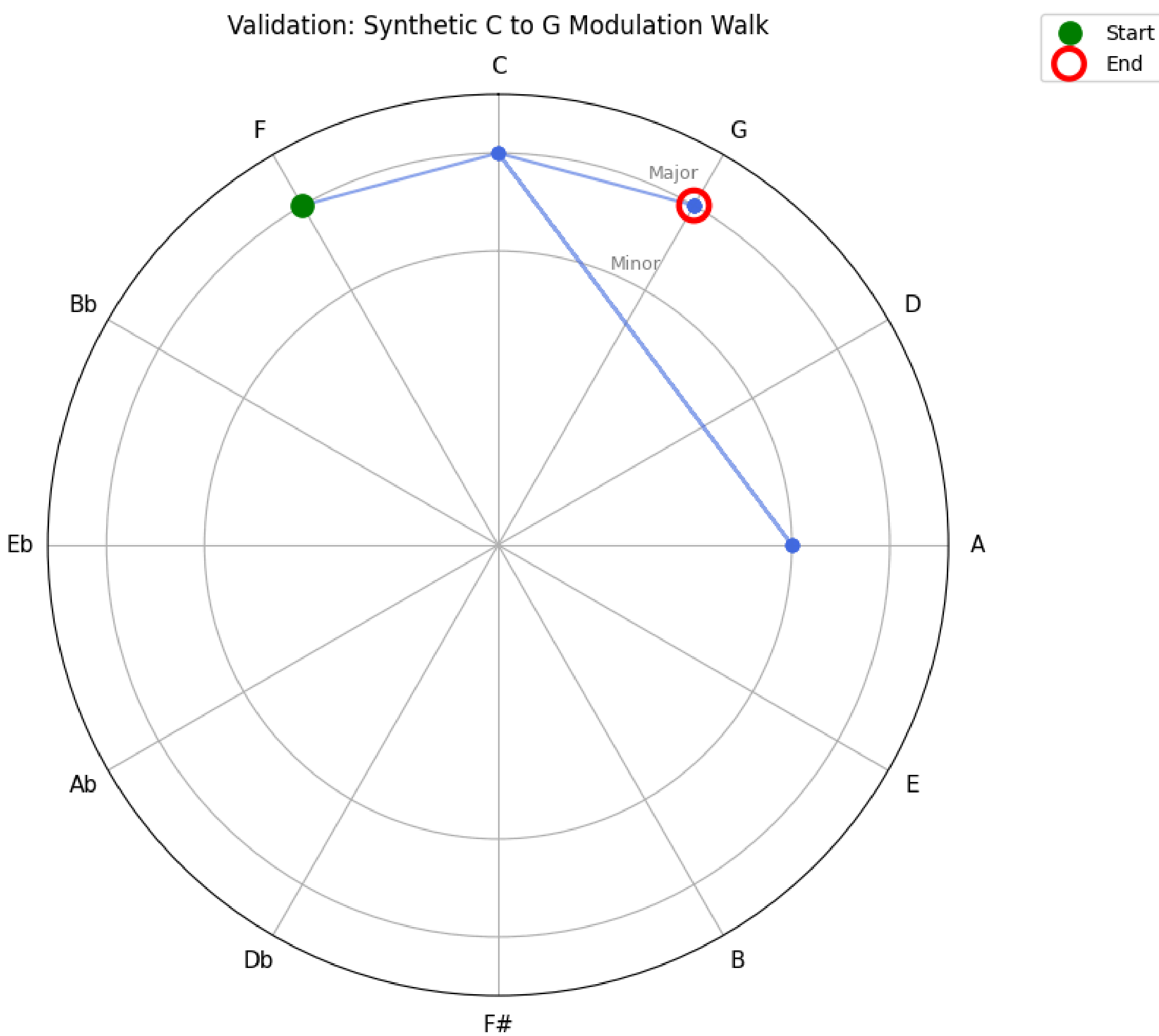

Figure 3. Validation: Synthetic C to G Tracking. Polar visualization of the model's most likely key estimate across the synthetic modulation sequence. The trajectory shows a shift from C major region toward G major region.

3.2. Real-World Test 1: Monophonic Melody

To assess the model's performance on real-world sequential music, we tested the pipeline using a monophonic MIDI melody of "Twinkle Twinkle Little Star". Because this simple music excerpt does not modulate, the expected global tonal center is C major throughout the sequence.

When visualized spatially on the Circle of Fifths (Figure 5), the trajectory remained closely clustered around C Major region. However, our analysis of the underlying probability distributions across time (Figure 4) revealed noticeable instability. Rather than producing a C Major centered signal, the model showed continuous competition among G major, F Major and E Minor.

This ambiguous result likely stems directly from our first representational mismatch. A single melodic note, like C, can belong to different diatonic triads, such as C Major, F Major, or A minor. When we forced the monophonic sequence into a vertical three-note triad template during preprocessing, the pipeline introduces additional ambiguity. The network's fluctuating probability distribution reflects the results of its systematic constraints: the triadic approximation is structurally insufficient to resolve the complex nature of horizontal sequenced melody.

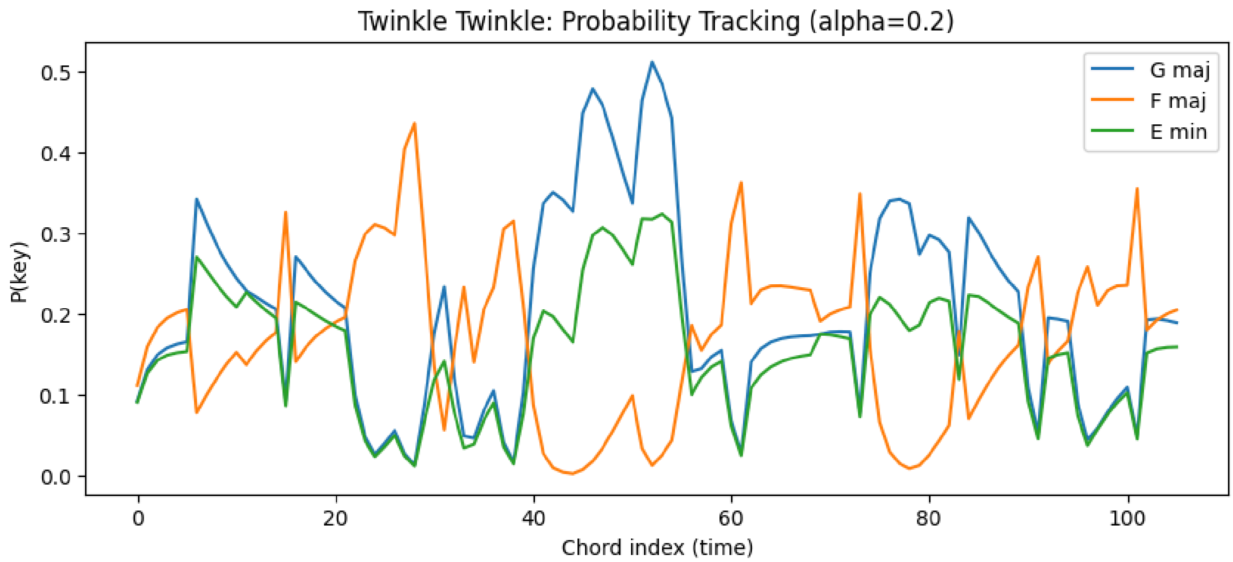

Figure 4. Real-World Test 1: Probability Tracking. Key probability estimates across the monophonic melody of "Twinkle Twinkle Little Star." The graph illustrates competing key hypotheses across time, especially among closely related keys.

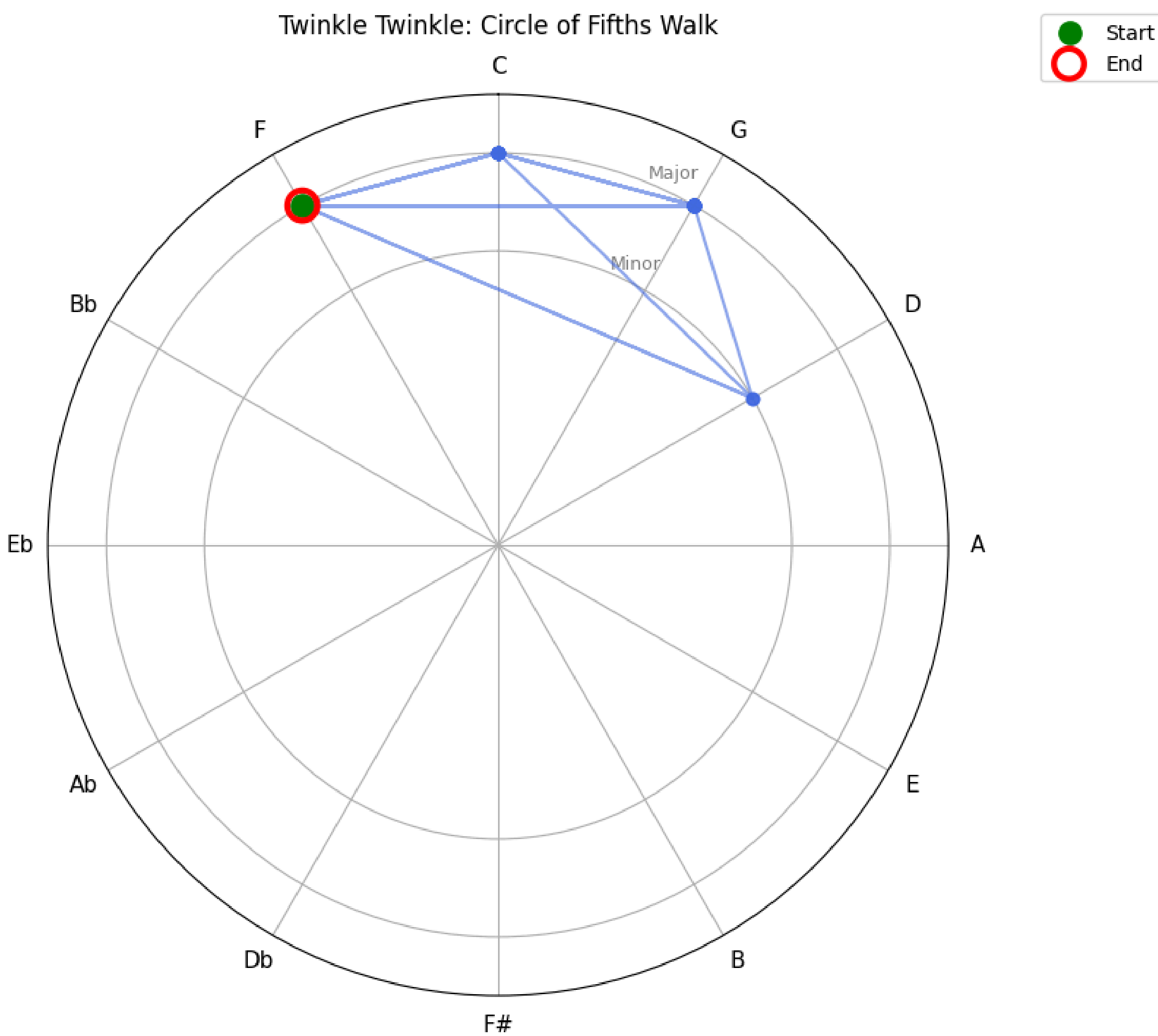

Figure 5. Real-World Test 1: Circle of Fifths Walk. Polar visualization of the model's key estimates for "Twinkle Twinkle Little Star." The trajectory remains clustered around closely related tonal regions, but shows spatial jumps caused by ambiguity in the triadic approximation.

3.3 The Ornamentation Collapse

To stress-test our pipeline's performance under conditions of high melodic complexity, we processed a MIDI recording of Mozart's 12 Variations on "Ah vous dirai-je, Maman" (the "Twinkle" theme). This composition, compares to the original style, introduces rapid sixteenth notes, trills, and frequent passing tones, which adds substantial amount of sequential information and corresponding increases of density.

Under the experimental condition (Method A: Triadic Forcing), the model revealed a near complete tonal collapse. As the EMA buffer integrated the rapid chromatic ornamentation over time, it simultaneously generated smeared and harmonically ambiguous frames. When the preprocessing algorithm forced these smeared acoustic frames into vertical three-note triads, the neural network produced highly inconsistent and volatile shifting patterns across the duration of the piece (Figure 6). This instability can also be observed directly in our extracted triadic representations. During the actively ornamented passages, the preprocessing stage repeatedly formed chromatic note clusters into unrelated major triads which produced output such as:

Extracted Chords (First 15):

C:maj | C:maj | C#:maj | C#:maj | C#:maj | C#:maj | C#:maj | C#:maj | C#:maj | C#:maj | C#:maj | C#:maj | C#:maj | C#:maj | C#:maj

Instead of maintaining a stable tonal center, the network began to produce abrupt transitions into musically implausible key regions that are also distant to each other. The Circle of Fifths visualizations (Figure 7) shows this failure spatially, as we observed large discontinuous jumps across the tonal space instead of coherent clustering around a stable key center.

These findings suggest that the failure does not necessarily originate from the neural network itself, but rather from the representational mismatch created by forcing rapidly and ornamented melodies into static and vertical triadic harmony structures.

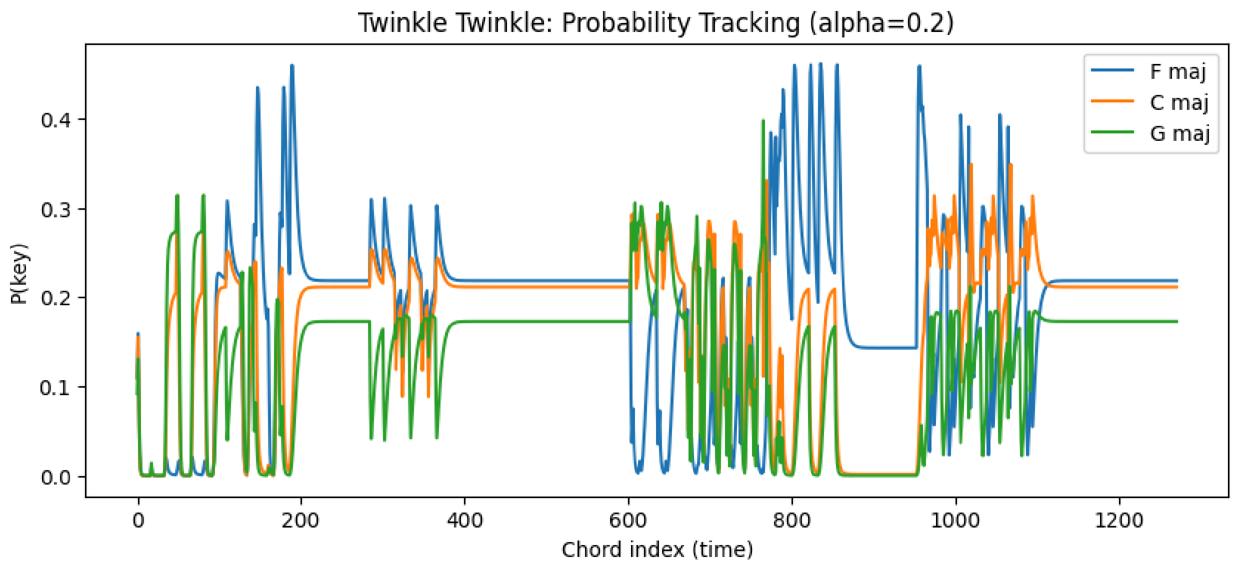

Figure 6. Real-World Test 2: Probability Tracking. Key probability estimates across Mozart's 12 variations. The distribution now becomes highly volatile under triadic forcing. The tonal inference is unstable.

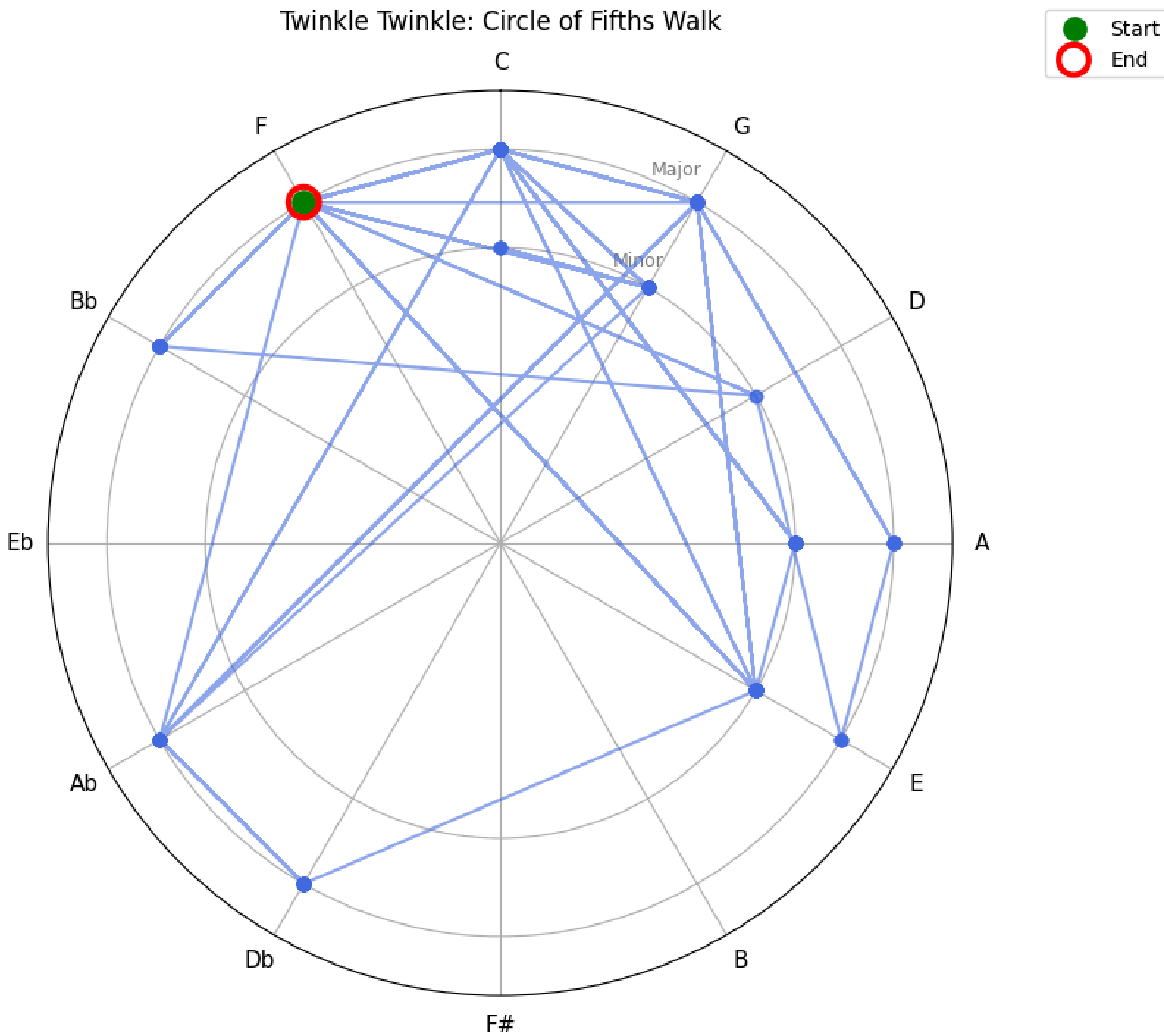

Figure 7. Real-World Test 2: The Ornamentation collapse. Polar visualization of the model's key estimates for Mozart's 12 Variations. The trajectory demonstrates severe drifting of the global tonal center caused by the forced triadic interpretation of rapid ornamentation.

3.4. The Pitch-Class Control

To determine whether the tonal collapse observed from the Mozart's 12 Variations originated from the ornamented audio itself or the representational mismatch from the model, we reanalyzed the same music piece using the the control baseline (Method B: Pitch-Class Distribution).

By bypassing both the triadic approximation and the neural network, this control method accumulated raw pitch-class frequencies over time and compared the resulting global distribution against fixed key templates. Unlike the triadic-forcing condition (Method A), the control baseline produced stable tonal estimates across the duration of the piece. The first 15 key predictions were all identified as C major:

Baseline Key Predictions (First 15):

C maj | C maj | C maj | C maj | C maj | C maj | C maj | C maj | C maj | C maj | C maj | C maj | C maj | C maj | C maj

These key predictions strongly suggest that the instability we observed under Method A did not originate from the audio input itself, but rather from the representational mismatch we proposed earlier. The triadic approximation fails to organize and interpret the rapid sequential ornamentation, whereas pitch-class accumulation remains robust to local variations and successfully captured the global tonal center.

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4. Discussion & Future Directions

4.1. The Representational Trade-off

The results of this study demonstrate that computational tonal inference is dependent on how auditory input is structurally represented over time. Our findings highlight the important representational trade-off in processing musical audio data.

The triadic representation (Method A) is highly sensitive to local harmonic structure and is capable of tracking dynamic shifts in tonal centers during clean chordal modulations. However, it becomes fragile when confronted with sequential and highly ornamented melodies. The forced compression of rapid passing tones into static vertical chord structures from Mozart's 12 variations generates unstable harmonic representations, which lead to systematic tonal instability. In contrast, the pitch-class baseline (Method B) remains robust to the melodic variation and successfully captures the global tonal center. However, this robustness was achieved at the cost of sensitivity to local harmonic movement and modulation.

These contrasting behaviors align well with existing cognitive theories of music perception: Human listeners likely combine both local harmonic information and broader melodic context, rather than solely relying on isolated chord structures (Palmer & Krumhansl, 1990). The contrasting failure modes observed in our two methods therefore suggest that the cognitive mechanisms responsible for processing vertical harmony and horizontal melody may rely on different but interacting representational systems.

4.2 Future Directions: Internalizing Time and Measuring Expectancy

Looking forward, the next step for our current computational modeling is to transition from externally added memory to internally learned memory states. Currently, auditory smoothing is imposed on the input data via the EMA buffer we added. Upgrading the architecture from the current feed-forward MLP to a Simple Recurrent Network (SRN), such as an Elman network (1990), would allow the model to maintain an internal temporal state (Defays, French, & Tillmann, 2023). This shift would allow the model to learn sequential dependencies naturally and learn temporal structure directly from the training data.

Once the network is capable of internalizing temporal structure, its predictive behavior could be evaluated using established paradigms from cognitive psychology. Specifically, applying Marilyn Boltz's (1993) partial-melody tasks would allow us to test the network's tonal prediction and expectancies. By strategically feeding the recurrent model incomplete melodic sequences, and analyzing its probability distributions at the exact moment of interruption, we can directly compare the model's implicit tonal expectations against human behavioral responses.

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5. Conclusion

In conclusion, our findings show that there is no single representational format capable of handling all aspects of tonal inference. While triadic models are excellent at extracting tonal centers from vertical harmony, they failed to maintain stable tonal inference when forced to process rapid and sequential musical notes. On the other hand, accumulating pitch classes information over time makes it easy to capture the global key of both simple and complex music excerpts, but this approach largely ignores local harmonic movement and chord changes.

These findings suggest that human music perception likely relies on multiple interacting cognitive systems rather than a single independent mechanism for decoding musical structure. As computational modeling moves toward more advanced architectures, such as recurrent networks capable of internalizing temporal structure, understanding how different musical representations interact with each other over time will become important for building models that more closely reflect true human music cognition.

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